



**SUBMITTED BY,
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Problem Statement

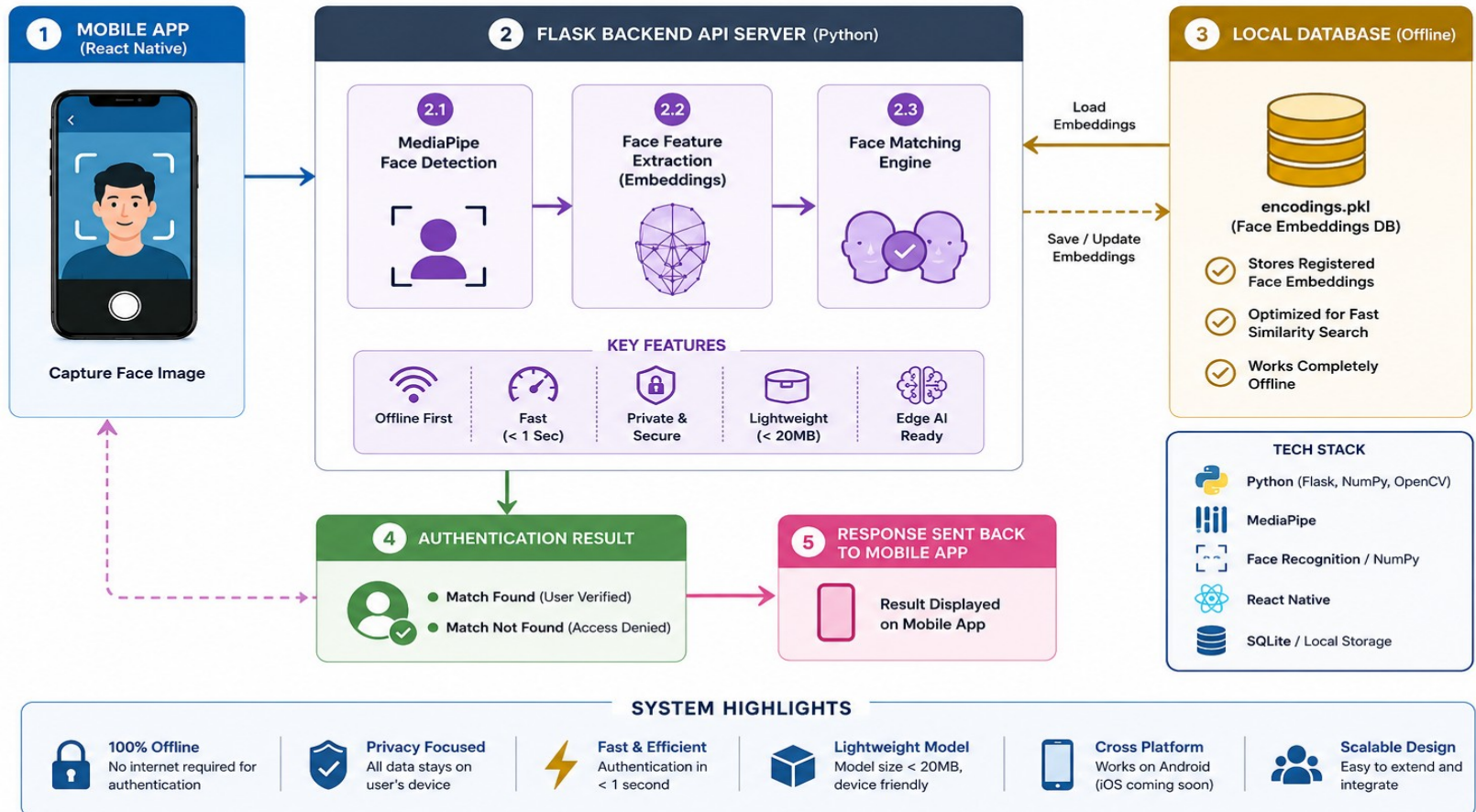
- Privacy risks in cloud-based face authentication
- Internet dependency
- High latency
- Poor usability in low-connectivity environments

Proposed Solution

- Fully offline-first architecture
- Local biometric processing
- No cloud dependency
- Privacy by design

System Architecture

Offline AI Face Authentication System Architecture



Face Recognition Pipeline

- Face enrollment
- Embedding generation
- Local encoding storage
- Authentication through similarity matching

Technology Stack

- React Native, Expo, Flask, Python, OpenCV, MediaPipe, NumPy

Future Scope

- Liveness Detection
- Encrypted Storage
- Multi-user Support

Real-World Impact

- Rural identity systems
- Campus authentication
- Offline access control
- Edge AI applications

Thank You